

Mason Shields

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Education

University of Michigan

Bachelor of Science in Computer Science

Ann Arbor, MI

Aug 2023-Dec 2026

- GPA: 3.5 /4.0
- Significant Courses:
 - EECS 388: Intro to Computer Security
 - EECS 485: Web Development and Infrastructure
 - EECS 493: User Interface Design
 - EECS 482: Operating Systems

Experience

Rocket Mortgage

Software Engineering Intern

Detroit, MI

May 2026-Aug 2026

- Redesigned an Angular workflow interface for automated mortgage origination, fixing state management bugs, leading to an increase of desired UI-first workflow creation usage by 50%
- Refactored legacy .NET schema generation endpoints into modular backend services, standardizing API structures to support automated origination pipelines as adoption grows
- Integrated a custom Claude Skill into the CI/CD pipeline to automatically update internal documentation sites directly from codebase changes during build runs, entirely reviving usage of said sites among a team of 20 engineers

Milieu Bio

Contract Software Engineer

Ann Arbor, MI

May 2025-Aug 2025

- Engineered an integration linking Google OAuth and our Supabase authentication backend framework, scaling role-based access seamlessly across hundreds of user accounts
- Integrated a full CI/CD pipeline into production website deployment using Github Actions and Cloudflare, entirely eliminating deployment coordination downtime

Kresge Hearing Research Institute

Undergraduate Research Software Developer

Ann Arbor, MI

Oct 2023-Present

- Developed computer vision analysis pipelines in Python using MediaPipe and OpenCV to quantify motor movement differences between Parkinson's patients and control subjects.
- Migrated local MATLAB data analysis scripts to a hosted cloud environment, enabling multi-user remote access for upcoming clinical trials.

Nebraska Innovation Labs

Software Development Intern

Omaha, NE

Jun 2024-Aug 2024

- Ported a company data table library from jQuery to React under guidance of a senior developer, greatly improving user control over data presentation
- Developed a greenfield React web application for commercial building management, building responsive dashboard interfaces and client-side routing.

Technologies

- **Programming Languages:** C#/.NET, Typescript, Python, C/C++, SQL, HTML/CSS, MATLAB, Java
- **Frameworks:** Angular, React.js, Laravel, Tailwind, SciPy/NumPy, Supabase, Selenium, Next.js
- **Technologies:** Linux Bash, Git/Github Version Control, Github Actions, Wireshark, AWS, Docker/Kubernetes